KAPASA MAKASA UNIVERSITY

ICT DEPARTMENT

CYS 110 TEST 2

17th MAY 2023

INSTRUCTIONS

Do not turn this page until you are told to do so

This Question paper contains 2 Questions, attempt all questions.

Write your name, student ID and Program clearly on the answer sheet.

Time allowed is 1hr 30 mnts hours only

Only calculators are allowed in the Exam room.

CamScanner

[QUESTION 1]

A. Using the RSA encryption algorithm, pick p = 11 and q = 7. Find a set of encryption/decryption keys e and d. (n,e) = (77, 7) and (n,d) = (77,43) **[10 Marks]** ✓

B. Using the private and public keys calculated above, encrypt the word **EXAM** and provide its cipher text. UUAJ **[10 Marks]** ✓

C. Your junior course mate asks you to help him decode the message **COHOBRGYRRYRITAPASHVTPFEE** he has just received from their CYS 110 lecturer. Use transposition cipher to decode and give him the answer Cryptography is for the brave **[2.5 Marks]** ✓

D. The three blocks of hexadecimal numbers **24 45 37, 64 44 A1** and **3B 22 09** need to be hushed in order to get a check sum so that the information they contain maintains its integrity. Calculate the checksum using hexadecimal addition. C3 AB E1 **[2.5 Marks]** ✓

[QUESTION 2]

A. You receive a message "**K, R, K, C, Q**" that was sent using the RSA public encryption key (21, 5). Decode this to find the original numeric and text message. BIBLE **[10 Marks]** ✓

B. Compare and contrast between Blowfish, Twofish and Serpent ciphers. **[5 Marks]** ✓

C. The MD5 hash algorithm of Ron Rivest is in wide use today. Give the four steps involved in creating the 128 bit hush. **[5 Marks]** ✓

D. Define Authentication and hence differentiate between endpoint authentication and Message Authentication by giving appropriate examples. **[5 Marks]** ✓

7 43

THE END AND ALL THE BEST

C B R
O R I
H G T
O Y A

C G
O
H
O
B
R

COHO
BRGY
RRYR

CRYPT
OGRAP
HYISF
ORTHE
BRAVE

C Y A F
O R S E
H I H E
O T V
B A T
R P P

CRYPTOGRAPHY
OGRA
PY

10+

CRYPTOGRAPHY

TEST 2

TERM 2

ANSWERS ONLY (MAKING KET)

QUESTION 1.

A. $n = p \times q$ $m = (p-1) \times (q-1)$ $e = 7$

$= 11 \times 7$ $= (11-1) \times (7-1)$ 2, 3, 5, (7), 11, 13, 17, 19, 29,

$= 77$ $= (10) \times (6)$ 31, 33, 37, 39, 41, 43 ....

$= 60$ ----- less than 77.

$\frac{ed - 1}{m}$ $\therefore d = 43$

$\frac{(7)(43) - 1}{60}$

$\frac{301 - 1}{60}$

$\frac{300}{60}$

$5, R = 0$

OR

$e = 11$ and $\frac{ed - 1}{m}$ $\therefore d = 11$

$\frac{(11)(11) - 1}{60}$

$\frac{121 - 1}{60}$

$120 - 1$

8. Public key $(n, e) = (77, 7)$
Private key $(n, d) = (77, 43)$

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

E = 5
X = 24
A = 1
M = 13

$C = m^e \bmod n$
$= 5^7 \bmod 77$
$= 47$
$= 47 - 26$
$= 21$
$S(21) = U$

$C = m^e \bmod n$
$= 24^7 \bmod 77$
$= 73$
$= 73 - 26$
$= 47 - 26$
$= 21$
$S(21) = U$

$C = m^e \bmod n$
$= 1^7 \bmod 77$
$= 1$
$S(1) = A$

$C = m^e \bmod n$
$= 13^7 \bmod 77$
$= 62$
$= 62 - 26$
$= 36 - 26$
$= 10$
$S(10) = J$

Plain text : E X A M
Cipher text : U U A J

OR

Public key $(n, e) = (77, 11)$
Private key $(n, d) = (77, 11)$

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$C = m^e \bmod 77$
$= 5^{11} \bmod 77$
$= 38$
$= 38 - 26$
$= 12$
$S(12) = L$

$C = m^e \bmod n$
$= 24^{11} \bmod 77$
$= 68$
$= 68 - 26$
$= 42 - 26$
$= 16$
$S(16) = P$

$C = m^e \bmod n$
$= 1^{11} \bmod 77$
$= 1$
$S(1) = A$

$C = m^e \bmod n$
$= 13^{11} \bmod 77$
$= 13$
$S(13) = M$

Plain text : E X A M
Cipher text : L P A M

PROOF :

For $e = 7$ and $d = 43$, plain text ? Cipher = J, 62
$m = c^d \bmod n$
$= 62^{43} \bmod 77$
$= 13$
$S(13) = M$
∴ plain text : M
Cipher text : J

For $e = 11$ and $d = 11$, plain text ? Cipher text = M, 13
$m = c^d \bmod n$
$= 13^{11} \bmod 77$
$= 13$
$S(13) = M$
∴ plain text : M
Cipher text : M

CamScanner

C. COHOBRGYRRYRITAPASHYTPFEE = 25 characters

matrix = 5 x 5

C R Y P T
O G R A P
H Y I S F
O R T H E
B R A Y E

∴ "Cryptography is for the brave".

D.

ASCII

24	45	37
64	44	A1
3B	22	09
C3	AB	E1

0	17 = 1.0625	19 = 1.1875
1	16	16
2	1 x 16 = 16	1 x 16 = 16
3	17 - 16 = 1, R	19 - 16 = 3, R
4		
5		
6		
7		
8		
9		
A (10)		
B (11)		
C (12)		
D (13)		
E (14)		
F (15)		

CamScanner

QUESTION 2

A. Public key $(n, e) = (21, 5)$

$p = 7$ $\quad m = (p-1) \times (q-1)$ $\quad \frac{ed-1}{m}$ $\quad \therefore d = 5$

$q = 3$ $\quad = (7-1) \times (3-1)$ $\quad \frac{(5)(5)-1}{12}$

$= 6 \times 2$ $\quad \frac{25-1}{12}$

$= 12$ $\quad \frac{24}{12} = 2, R = 0$

Private key

$(n, d) = (21, 5)$

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

$K = 11$ $\quad m = c^d \bmod n$ $\quad m = c^d \bmod n$

$R = 18$ $\quad = 11^5 \bmod 21$ $\quad = 18^5 \bmod 21$

$K = 11$ $\quad = 2$ $\quad = 9$

$C = 3$ $\quad S(2) = B$ $\quad S(9) = I$

$Q = 17$

$m = c^d \bmod n$ $\quad m = c^d \bmod n$

$= 17^5 \bmod 21$ $\quad = 3^5 \bmod 21$

$= 5$ $\quad = 12$

$S(5) = E$ $\quad S(12) = L$

∴

Plain text : B I B L E

Cipher text : K R K C Q

C. STEP 1, padding - adding a 1 followed by enough 0s to satisfy certain message conditions.
STEP 2, Appending - Apend a 64-word block before padding.
STEP 3, Initialize the accumulator.
STEP 4, Looping - a 16-word block in 4-rounds.

D. Authentication - is proving user identity, e.g putting a pin when unlocking your phone.
Endpoint authentication - is the verifying of entities that are communicating over a computer network, e.g in ap 4.0, a user and server verifies each others identity, for example a user provides a user name and password, then the server sends a nonce so that the user proves his/her identity to the sever.
Message Authentication - is the verification that proves that a message is legitimate and hasn't lost its integrity, for example the use of hash functions such as MD5 and SHA-1, $x = y$ or $H(m) = h$ proves a message wasn't altered or changed during its transmission.

THE END AND ALL THE BEST

~~Tisola~~

Kipeu Dc